Report on Visualizations

Assignment 3

Dataset link - https://www.kaggle.com/datasets/mohamedmagdy511/olist-brazilian-e-commerce

This document lists every visual in the dashboard, organized page-by-page, with chart type, data fields used, table connections, and what each visual represents.

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1. Executive Overview

1. **Card: Total Orders**
    - **Measure:** Total Orders = COUNTROWS(Orders)
    - **Source Table:** Orders
    - **Represents:** Total count of orders.
2. **Card: Avg Delivery Days**
    - **Measure:** Avg Delivery Days = AVERAGE(DATEDIFF(Orders[order_purchase_timestamp], Orders[order_delivered_customer_date], DAY))
    - **Source Table:** Orders
    - **Represents:** Average days between purchase and delivery.
3. **Card: Total GMV**
    - **Measure:** Total GMV = SUM('Order Items'[price])
    - **Source Table:** Order Items
    - **Represents:** Sum of all line-item prices.
4. **Card: Avg Review Score**
    - **Measure:** Avg Review Score = AVERAGE(Reviews[review_score])
    - **Source Table:** Reviews
    - **Represents:** Mean customer rating.
5. **Card: Avg Basket Value**
    - **Measure:** Avg Basket Value = AVERAGEX(VALUES(Orders[order_id]), CALCULATE(SUM('Order Items'[price])))
    - **Source Tables:** Orders → Order Items
    - **Represents:** Average order total.
6. **Combo Chart: Orders & GMV by Month**

- **Chart Type:** Line & Column
- **Axis:** Calendar[Month]
- **Column Value:** Total Orders
- **Line Value:** Total GMV
- **Source Tables:** Orders → Calendar; Orders → Order Items
- **Represents:** Monthly order volume vs revenue.

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2. Time Trend

1. **Combo Chart: Orders & GMV by Month**
   - *Same as Executive Overview #6*
2. **100% Stacked Bar: Orders by Quarter & Payment Type**
   - **Chart Type:** 100% Stacked Column
   - **Axis:** Calendar[Quarter]
   - **Legend:** Order Payments[payment_type]
   - **Value:** Total Orders
   - **Tables:** Orders → Calendar; Orders → Order Payments
   - **Represents:** Payment mix share per quarter.
3. **Bar: Orders by Day of Month**
   - **Axis:** Calendar[Day]
   - **Value:** Total Orders
   - **Tables:** Orders → Calendar
   - **Represents:** Order counts by calendar day.
4. **Bar: Orders by Weekday**
   - **Axis:** Orders[Purchase Day]
   - **Value:** Total Orders
   - **Table:** Orders
   - **Represents:** Volume by day of week.

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3. Category Performance

1. **Bar: Orders per Category**
    - **Axis:** Order Items[Product Category EN]
    - **Value:** Orders per Category = DISTINCTCOUNT(Orders[order_id])
    - **Tables:** Order Items → Orders
    - **Represents:** Order counts per product category.
2. **Stacked Column: GMV by Category & Quarter**
    - **Axis:** Order Items[Product Category EN]
    - **Legend:** Calendar[Quarter]
    - **Value:** Total GMV
    - **Tables:** Order Items → Orders → Calendar
    - **Represents:** Revenue distribution by category over quarters.

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4. Customer Behaviour

1. **Stacked Bar: Orders by City & Payment Type**
    - **Axis:** Orders[customer_city]
    - **Legend:** Order Payments[payment_type]
    - **Value:** Total Orders
    - **Tables:** Orders → Order Payments
    - **Represents:** City-level orders broken down by payment method.
2. **Donut: Avg Basket Value by Day of Month**
    - **Legend:** Orders[Day]
    - **Value:** Avg Basket Value
    - **Tables:** Orders → Order Items
    - **Represents:** Variation of average order value across month days.

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5. Geography

1. **Bubble Map: Orders & Basket by City**
    - **Latitude:** Geolocation[geolocation_lat]
    - **Longitude:** Geolocation[geolocation_lng]

- **Size:** Total Orders
- **Color:** Geolocation[geolocation_state]
- **Tooltip:** Avg Basket Value
- **Tables:** Orders → Customers → Geolocation; Orders → Order Items
- **Represents:** Geographic order volume with basket size.

2. **Choropleth: Orders by State**
    - **Location:** Geolocation[geolocation_state]
    - **Color:** Total Orders
    - **Tables:** Orders → Customers → Geolocation
    - **Represents:** State-level order distribution.

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6. Delivery Logistics

1. **Line: Avg Delivery vs Estimated by Month**
    - **Axis:** Calendar[Month]
    - **Lines:** Avg Delivery Days, Avg Estimated Days
    - **Table:** Orders; Calendar
    - **Represents:** Actual vs promised delivery durations monthly.
2. **Sunburst: Late Delivery % by Quarter & Month**
    - **Inner:** Calendar[Quarter]
    - **Outer:** Calendar[Month]
    - **Value:** Late Delivery %
    - **Tables:** Orders; Calendar
    - **Represents:** Late delivery proportions by period.
3. **Bar: Late Delivery % by Seller**
    - **Axis:** Order Items[seller_id]
    - **Value:** Late Delivery %
    - **Tables:** Order Items → Orders
    - **Represents:** Seller-specific late rates.

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7. Payment & Fraud Check

1. **Matrix: Orders by Payment & Installments**
    - **Rows:** Order Payments[payment_type]
    - **Columns:** Order Payments[payment_installments]
    - **Values:** Total Orders
    - **Tables:** Orders → Order Payments
    - **Represents:** Order counts by payment method and installment count.
2. **Line: Total Orders by Payment Type**
    - **Axis:** Order Payments[payment_type]
    - **Value:** Total Orders
    - **Table:** Order Payments; Orders
    - **Represents:** Order volume per payment method.
3. **Pie: Installment Share %**
    - **Legend:** Order Payments[payment_installments]
    - **Value:** Instalment Share %
    - **Tables:** Orders → Order Payments
    - **Represents:** Order share by installment count.
4. **Bar: Total Orders by Payment**
    - **Axis:** Order Payments[payment_type]
    - **Value:** Total Orders
    - **Tables:** Orders → Order Payments
    - **Represents:** Payment-type order counts.

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8. Reviews & Satisfaction

1. **Card: Avg Review Score**
    - **Measure:** Avg Review Score
    - **Table:** Reviews
    - **Represents:** Overall customer rating.

2. **Bar: Total Reviews by Score**
    - **Axis:** Reviews[review_score]
    - **Value:** Total Reviews
    - **Table:** Reviews
    - **Represents:** Distribution of review ratings.
3. **Donut: Total Reviews by Score**
    - **Legend:** Reviews[review_score]
    - **Value:** Total Reviews
    - **Table:** Reviews
    - **Represents:** Proportional review breakdown by rating.